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Measure Theory

1.1 Outer Measure

Definition 1.1.1 ► Outer Measure

A function φ defined for every subset A of an arbitrary set X is called an **outer measure** on X if the following conditions are satisfied:

- (i) $\varphi(\emptyset) = 0$,
- (ii) $0 \leq \varphi(A) \leq \infty$ whenever $A \subset X$,
- (iii) $\varphi(A_1) \leq \varphi(A_2)$ whenever $A_1 \subset A_2$,
- (iv) $\varphi(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \varphi(A_i)$ for every countable collection of sets $\{A_i\}$ in X .

Example 1.1.2 ► Outer Measure

- (i) In an arbitrary set X , define $\varphi(A) = 1$ if A is nonempty and $\varphi(\emptyset) = 0$.
- (ii) Let $\varphi(A)$ be the number (possibly infinite) of points in A .
- (iii) Let $\varphi(A) = \begin{cases} 0 & \text{if card } A \leq \aleph_0, \\ 1 & \text{if card } A > \aleph_0. \end{cases}$
- (iv) If X is a metric space, fix $\varepsilon > 0$. Let $\varphi(A)$ be the smallest number of balls of radius ε that cover A .
- (v) Select a fixed x_0 in an arbitrary set X , and let

$$\varphi(A) = \begin{cases} 0 & \text{if } x_0 \notin A, \\ 1 & \text{if } x_0 \in A; \end{cases}$$

φ is called the **Dirac measure** concentrated at x_0 .

Definition 1.1.3

Let φ be an outer measure on a set X . A set $E \subset X$ is called **φ -measurable** if

$$\varphi(A) = \varphi(A \cap E) + \varphi(A - E)$$

for every set $A \subset X$. In view of property (iv) above, observe that φ -measurability requires only

$$\varphi(A) \geq \varphi(A \cap E) + \varphi(A - E).$$

Note.

也就是说, E 是可测的, 当且仅当任意集合 A 都可以被 E 按外测度被良好地拆解为 $A \cap E$ 和 $A \setminus E$.

Lemma 1.1.4

A set $E \subset X$ is φ -measurable if and only if

$$\varphi(P \cup Q) = \varphi(P) + \varphi(Q)$$

for all sets P and Q such that $P \subset E$ and $Q \subset E^c$.

Note.

也就是说 E 可测, 当且仅当落在 E 的内, 外的集合在外测度意义下是无关的.

Proof. For Necessity, if E is φ -measurable, then for all such P, Q ,

$$\varphi(P \cup Q) = \varphi((P \cup Q) \cap E) + \varphi((P \cup Q) - E) = \varphi(P) + \varphi(Q)$$

For sufficiency, let $P = A \cap E$, $Q = A \cap E^c = A \setminus E$. Then

$$\varphi(A) = \varphi(P \cup Q) = \varphi(P) + \varphi(Q) = \varphi(A \cap E) + \varphi(A \setminus E)$$

□

Theorem 1.1.5

Suppose φ is an outer measure on an arbitrary set X . Then the following four statements hold:

- (i) E is φ -measurable whenever $\varphi(E) = 0$.
- (ii) $\emptyset$ and X are φ -measurable.
- (iii) $E_1 - E_2$ is φ -measurable whenever E_1 and E_2 are φ -measurable.
- (iv) If $\{E_i\}$ is a **countable collection of disjoint φ -measurable sets**, then

$\bigcup_{i=1}^{\infty} E_i$ is φ -measurable and

$$\varphi\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \varphi(E_i).$$

More generally, if $A \subset X$ is an arbitrary set, then

$$\varphi(A) = \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c),$$

where

$$S = \bigcup_{i=1}^{\infty} E_i.$$

Proof. 1. Take any $A \subseteq X$. Since $A \cap E \subseteq E$, $\varphi(A \cap E) = 0$. Thus

$$\varphi(A \cap E) + \varphi(A \cap E^c) = \varphi(A \cap E^c) \leq \varphi(A)$$

2. Follows immediately from lemma 1.1.4.

3. Take any P, Q such that $P \subseteq (E_1 \setminus E_2) = E_1 \cap E_2^c$, $Q \subseteq (E_1 \setminus E_2)^c = E_1^c \cup E_2$.

$$\varphi(Q) = \varphi(Q \cap E_2) + \varphi(Q - E_2)$$

Note that $Q - E_2 \subseteq E_1^c$, $Q \cap E_2 \subseteq E_2$. Since $P \subseteq E_1$, the measurability of E_1 implies that

$$\varphi(P) + \varphi(Q - E_2) = \varphi(P \cup (Q - E_2))$$

Note that $P \cup (Q - E_2) \subseteq E_2^c$. Since $Q \cap E_2 \subseteq E_2$, we have

$$\varphi(P \cup (Q - E_2)) + \varphi(Q \cap E_2) = \varphi(P \cup Q)$$

Finally, we have

$$\begin{aligned} \varphi(P) + \varphi(Q) &= \varphi(P) + \varphi(Q \cap E_2) + \varphi(Q - E_2) \\ &= \varphi(P \cup (Q - E_2)) + \varphi(Q \cap E_2) \\ &= \varphi(P \cup Q) \end{aligned}$$

4. Let $S_k = \bigcup_{i=1}^k E_i$. We proceed by finite induction and first note that the result is obviously true for $k = 1$. For $k > 1$ assume that S_k is φ -measurable and that

$$\varphi(A) \geq \sum_{i=1}^k \varphi(A \cap E_i) + \varphi(A \cap S_k^c), \quad \forall A \subseteq X$$

Then

$$\begin{aligned} \varphi(A) &= \varphi(A \cap E_{k+1}) + \varphi(A \cap E_{k+1}^c) \\ &= \varphi(A \cap E_{k+1}) + \varphi(A \cap E_{k+1}^c \cap S_k^c) + \varphi(A \cap E_{k+1}^c \cap S_k) \\ &= \varphi(A \cap E_{k+1}) + \varphi(A \cap S_{k+1}^c) + \varphi(A \cap S_k) \end{aligned}$$

Replace A by $A \cap S_k$, we have

$$\varphi(A \cap S_k) \geq \sum_{i=1}^k \varphi(A \cap S_k \cap E_i) + \varphi(A \cap S_k \cap S_k^c) = \sum_{i=1}^k \varphi(A \cap E_i)$$

Thus

$$\varphi(A) \geq \sum_{i=1}^{k+1} \varphi(A \cap E_{k+1}) + \varphi(A \cap S_{k+1}^c)$$

Then the subadditivity gives that

$$\varphi(A) \geq \varphi(A \cap S_{k+1}) + \varphi(A \cap S_{k+1}^c)$$

this, in turn, implies that S_{k+1} is φ -measurable. Since we now know for every set $A \subseteq X$ and for all positive integers k

$$\varphi(A) \geq \sum_{i=1}^k \varphi(A \cap E_i) + \varphi(A \cap S_k^c)$$

We have

$$\varphi(A) \geq \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c) \geq \varphi(A \cap S) + \varphi(A \cap S^c)$$

where $S = \bigcup_{i=1}^{\infty} E_k$. Thus S is measurable. The subadditivity gives that

$$\varphi(A) \leq \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c)$$

Finally, we have

$$\varphi(A) = \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c)$$

Replace A by S , we can get

$$\varphi(S) = \sum_{i=1}^{\infty} \varphi(E_i)$$

□

Lemma 1.1.6

Let $\{E_i\}$ be a sequence of arbitrary sets. Then there exists a sequence of disjoint sets $\{A_i\}$ such that each $A_i \subset E_i$ and

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i.$$

If each E_i is φ -measurable, and so is A_i .

Proof. Let $S_k = \bigcup_{i=1}^k E_i$. Let $A_k = S_k \setminus S_{k-1}$, $k \geq 1$, where $S_0 = \emptyset$. Then

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} (S_k \setminus S_{k-1}) = \bigcup_{i=1}^{\infty} A_i.$$

If each E_i is φ -measurable. $E_{i+1} \cup E_i = (E_{i+1} - E_i) \cup E_i$ is measurable follows from the (ii), (iii) of the above theorem. Inductively, every $S_k = (S_{k-1} \setminus E_k) \cup E_k$ is measurable. Then each A_k is measurable. □

Theorem 1.1.7

If $\{E_i\}$ is a sequence of φ -*measurable* sets in X , then $\cup_{i=1}^{\infty} E_i$ and $\cap_{i=1}^{\infty} E_i$ are φ -*measurable*.

Proof. By the lemma above, there exists a sequence of disjoint φ -measurable sets $\{A_i\}$ such that each $A_i \subseteq E_i$, and

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i$$

It follows from Theorem 1.1.5 that $\bigcup_{i=1}^{\infty} A_i$ is measurable, then $\bigcup_{i=1}^{\infty} E_i$ is as well. Since X is measurable, $E_i^c = X \setminus E_i$ is measurable, then $\bigcup_{i=1}^{\infty} E_i^c$ is measurable. Thus

$$\bigcap_{i=1}^{\infty} E_i = X \setminus \bigcup_{i=1}^{\infty} E_i^c$$

is measurable. □

Definition 1.1.8

A nonempty collection Σ of sets E satisfying the following two conditions is called a σ -*algebra*:

- (i) if $E \in \Sigma$, then $E^c \in \Sigma$;
- (ii) $\cup_{i=1}^{\infty} E_i \in \Sigma$ if each E_i is in Σ .

Definition 1.1.9

In a topological space, the elements of the smallest σ -algebra that contains all open sets are called **Borel sets**. The term “smallest” is taken in the sense of inclusion.

Proof. We need to show that the “smallest” exists. □

Corollary 1.1.10

If φ is an *outer measure* on an arbitrary set X , then the class of φ -measurable sets forms a σ -*algebra*.

Proof. Follows immediately from Theorem 1.1.5 and Theorem 1.1.7. □

Corollary 1.1.11

Suppose φ is an outer measure on X and $\{E_i\}$ a countable collection of φ -measurable sets.

(i) If $E_1 \subset E_2$ with $\varphi(E_1) < \infty$, then

$$\varphi(E_2 \setminus E_1) = \varphi(E_2) - \varphi(E_1).$$

(ii) (Countable additivity) If $\{E_i\}$ is a disjoint sequence of sets, then

$$\varphi\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \varphi(E_i).$$

(iii) (Continuity from the left) If $\{E_i\}$ is an increasing sequence of sets, that is, if $E_i \subset E_{i+1}$ for each i , then

$$\varphi\left(\bigcup_{i=1}^{\infty} E_i\right) = \varphi\left(\lim_{i \rightarrow \infty} E_i\right) = \lim_{i \rightarrow \infty} \varphi(E_i).$$

(iv) (Continuity from the right) If $\{E_i\}$ is a decreasing sequence of sets, that is, if $E_i \supset E_{i+1}$ for each i , and if $\varphi(E_{i_0}) < \infty$ for some i_0 , then

$$\varphi\left(\bigcap_{i=1}^{\infty} E_i\right) = \varphi\left(\lim_{i \rightarrow \infty} E_i\right) = \lim_{i \rightarrow \infty} \varphi(E_i).$$

(v) If $\{E_i\}$ is any sequence of φ -measurable sets, then

$$\varphi\left(\liminf_{i \rightarrow \infty} E_i\right) \leq \liminf_{i \rightarrow \infty} \varphi(E_i).$$

(vi) If

$$\varphi\left(\bigcup_{i=i_0}^{\infty} E_i\right) < \infty$$

for some positive integer i_0 , then

$$\varphi\left(\limsup_{i \rightarrow \infty} E_i\right) \geq \limsup_{i \rightarrow \infty} \varphi(E_i).$$

Proof. (v) Let $A_j = \bigcap_{k=j}^{\infty} E_k$, then

$$\liminf_{i \rightarrow \infty} E_i = \lim_{j \rightarrow \infty} A_j = \bigcup_{j=1}^{\infty} A_j$$

It follows from (iv) that

$$\varphi\left(\liminf_{i \rightarrow \infty} E_i\right) = \lim_{j \rightarrow \infty} \varphi(A_j)$$

Since $A_j \subseteq E_j$, we have $\varphi(A_j) \leq \varphi(E_j)$, then

$$\lim_{j \rightarrow \infty} \varphi(A_j) \leq \liminf_{i \rightarrow \infty} \varphi(E_i)$$

Thus

$$\varphi\left(\liminf_{i \rightarrow \infty} E_i\right) \leq \liminf_{i \rightarrow \infty} \varphi(E_i)$$

□

Definition 1.1.12

- An outer measure φ on a topological space X is called a **Borel outer measure** if all Borel sets are φ -measurable.
- A Borel outer measure is **finite** if $\varphi(X)$ is finite.
- An outer measure φ on a set X is called **regular** if for each $A \subseteq X$ there exists a φ -measurable set $B \supseteq A$ such that $\varphi(B) = \varphi(A)$.
- A **Borel regular outer measure** is a Borel set B such that $\varphi(B) = \varphi(A)$.
- A **Radon outer measure** is a Borel regular outer measure that is finite on compact sets.

Exercises for Outer Measure

Exercise 1.1.13

If X is a metric space, fix $\varepsilon > 0$. Let $\varphi(A)$ be the smallest number of balls of radius ε that cover A .

Let $\varphi_\varepsilon(A) := \varphi(A)$ denote the dependence on ε , and define

$$\psi(A) := \lim_{\varepsilon \rightarrow 0} \varphi_\varepsilon(A)$$

What is $\psi(A)$ and what are the corresponding ψ -measurable sets.

Proof. If there are only finite points in A . Let

$$l := \min \{d(x, y) : x, y \in A, x \neq y\}$$

Then for any $x, y \in A, x \neq y$. If B_1, B_2 are ball of radius $\frac{l}{2}$ such that $x \in B_1, y \in B_2$. Then $B_1 \cap B_2 = \emptyset$. But each point in A must covered by one of such ball, which shows that $\psi(A) \geq |A|$. Since it is obvious that $\psi(A) \leq |A|$, then $\psi(A) = |A|$.

Now suppose that A is a infinite set. If $\psi(A) < \infty$, say $\psi(A) = n$. Then for any sufficiently small ε , there are balls $B_1, \dots, B_n$ with radius ε cover A . Take any $n + 1$ different points $p_1, \dots, p_{n+1}$ of A . Let $\varepsilon = \frac{1}{2} \min \{d(p_i, p_j) : i \neq j\}$, then $B_1, \dots, B_n$ cannot cover these $n + 1$ points, thus cannot cover A , which is a contradiction. Thus we have $\psi(A) = \infty$.

We find that ψ is just the counting measure on X . Take any $E \subseteq X$, and take any $A \subseteq X$, if A is finite, then $A \cap E, A - E$ are as well, and $|A| = |A \cap E| + |A - E|$, that is $\psi(A) = \psi(A \cap E) + \psi(A - E)$. If A is infinite, then one of the $A \cap E$ and $A - E$ is as well. Then

$$\psi(A) = \infty = \psi(A \cap E) + \psi(A - E)$$

The above shows that E is ψ -measurable. Thus all subset of X are ψ -measurable. $\square$

Exercise 1.1.14

It was shown that $\varphi(E_2 \setminus E_1) = \varphi(E_2) - \varphi(E_1)$ if $E_1 \subseteq E_2$ are φ -measurable with $\varphi(E_1) < \infty$. Prove that this result remains true if E_2 is not assumed to be φ -measurable.

Proof. Since E_1 is measurable,

$$\varphi(E_2) = \varphi(E_2 \cap E_1) + \varphi(E_2 - E_1)$$

but

$$\varphi(E_2 \cap E_1) = \varphi(E_1)$$

which completes the proof. □

1.2 Carathéodory Outer Measure

Definition 1.2.1 ▶ Carathéodory outer measure

An outer measure φ defined on a metric space (X, ρ) is called a **Carathéodory outer measure** if

$$\varphi(A \cup B) = \varphi(A) + \varphi(B)$$

whenever A, B are arbitrary subsets of X with $d(A, B) > 0$. Where

$$d(A, B) := \inf\{\rho(a, b) : a \in A, b \in B\}$$

Theorem 1.2.2

If φ is a Carathéodory outer measure on a metric space X , then all closed sets are φ -measurable.

Proof sketch.

用一系列集合近似代替 C 是标准的想法. 证明的核心要素有: 1. C 是闭集, 使得按距离向外逼近的行为是良好的. 2. 距离这个属性与 Carathéodory 条件是相适应的. 此外, 将命题转化为极限

$$\lim_{i \rightarrow \infty} \varphi(A \setminus C_i) = \varphi(A \setminus C)$$

是标准的流程. 要证明该极限成立, 则需要一定的技巧. 这里将 $A \setminus C_i$ 和 $A \setminus C$ 相差的中间部分进行细分量化, 将原极限转化为

$$\lim_{i \rightarrow \infty} \sum_{j=i}^{\infty} \varphi(T_j)$$

的极限问题, 从而转化为级数的收敛性问题. 将集合隔一个分组, 变成两组两两相隔正距离的集合, 以便套用 Carathéodory 条件.

Proof. Let $C \subseteq X$ be any closed sets. Take $A \subseteq X$, only need to show that

$$\varphi(A) \geq \varphi(A \cap C) + \varphi(A \setminus C)$$

We assume that $\varphi(A) < \infty$. Define

$$C_i := \left\{ x : d(x, C) \leq \frac{1}{i} \right\}$$

Since $d(A \setminus C_i, A \cap C) > 0$,

$$\varphi(A) \geq \varphi((A \setminus C_i) \cup (A \cap C)) = \varphi(A \setminus C_i) + \varphi(A \cap C)$$

Only need to show that

$$\lim_{i \rightarrow \infty} \varphi(A \setminus C_i) = \varphi(A \setminus C)$$

Define

$$T_i := \left\{ x \in A : \frac{1}{i+1} < d(x, C) \leq \frac{1}{i} \right\}$$

, then

$$A \setminus C \subseteq (A \setminus C_i) \cup \bigcup_{j=i}^{\infty} T_j$$

Thus

$$\varphi(A \setminus C) \leq \varphi(A \setminus C_i) + \sum_{j=i}^{\infty} \varphi(T_j)$$

Once we have $\sum_{j=1}^{\infty} \varphi(T_j) < \infty$, then $\lim_{i \rightarrow \infty} \sum_{j=i}^{\infty} \varphi(T_j) = 0$, the conclusion follows from it. To show this, note that $d(T_i, T_j) > 0$ whenever $|i - j| \geq 2$. Then it by induction follows from Carathéodory condition that

$$\sum_{j=1}^{\infty} \varphi(T_{2j}) \leq \varphi(A) < \infty$$

$$\sum_{j=1}^{\infty} \varphi(T_{2j-1}) \leq \varphi(A) < \infty$$

Thus $\sum_{j=1}^{\infty} \varphi(T_j) < \infty$, which completes the proof. □

Theorem 1.2.3

Suppose $\mathcal{F}$ is a family of subsets of a topological space X that contains all open and all closed subsets of X . Suppose also that $\mathcal{F}$ is closed under countable unions and countable intersections. Then $\mathcal{F}$ contains all Borel sets; that is, $\mathcal{B} \subset \mathcal{F}$.

Note.

可以将包含 $\mathcal{B}$ 描述成对以下三种集合的封闭性: 1. 开集 2. 补集 3. 可数并. 如果我们事先没有对补集的封闭性, 可以通过加强 1.3. 的条件, 使得当我们强制使得补集封闭后, 得到的新集族依旧满足 1.3. 具体地, 我们希望原集族对于开集的补集, 也就是闭集也封闭; 对于可数并的补集, 也就是可数交也封闭.

Proof. Let

$$\mathcal{H} = \mathcal{F} \cap \{A : A^c \in \mathcal{F}\}$$

Observe that $\mathcal{H}$ contains all closed sets. Moreover, it is easily seen that $\mathcal{H}$ is closed under complementation and countable unions. Thus $\mathcal{H}$ is a σ -algebra that contains the open sets and therefore contains all Borel sets. $\square$

Theorem 1.2.4

If φ is a Carathéodory outer measure on a metric space X , then the Borel sets of X are φ -measurable.

Proof. Since the collection of φ -measurable sets is itself a σ -algebra, it is closed under complements and countable union and countable intersection. Theorem 1.2.2 gives that φ -measurable sets contains closed sets, thus contains open sets as well. Then the conclusion follows from Theorem 1.2.3. $\square$

Exercises for this section

Exercise 1.2.5 ▶ Smallest σ -algebra containing open sets

Prove that in every topological space X , there exists a smallest σ -algebra that contains all open sets in X . That is, prove that there is a σ -algebra Σ that contains all open sets and has the property that if Σ_1 is another σ -algebra containing all open sets, then $\Sigma \subset \Sigma_1$. In particular, for $X = \mathbb{R}^n$, note that there is a smallest σ -algebra that contains all the closed sets in $\mathbb{R}^n$.

Proof. Let

$$\Sigma := \bigcap_{\mathcal{M}_\alpha \text{ is a } \sigma \text{ algebra containing } X} \mathcal{M}_\alpha$$

Then it is obvious that Σ is closed under complement, countable union, and contains all open sets. Thus Σ is a σ -algebra containing all open sets. $\square$

Exercise 1.2.6

In a topological space X the family of Borel sets, $\mathcal{B}$, is by definition the σ -algebra generated by the closed sets. The method below is another way of describing the Borel sets using transfinite induction. You are to fill in the necessary steps:

1. For an arbitrary family $\mathcal{F}$ of sets, let

$$\mathcal{F}^* = \left\{ \bigcup_{k=1}^{\infty} E_k : \text{where either } E_i \in \mathcal{F} \text{ or } \tilde{E}_i \in \mathcal{F} \text{ for all } i \in \mathbb{N} \right\}.$$

Let Ω denote the smallest uncountable ordinal. We will use transfinite induction to define a family $\mathcal{E}_\alpha$ for each $\alpha < \Omega$.

2. Let $\mathcal{E}_0 :=$ all closed sets $:= \mathcal{K}$. Now choose $\alpha < \Omega$ and assume that $\mathcal{E}_\beta$ has been defined for each β such that $0 \leq \beta < \alpha$. Define

$$\mathcal{E}_\alpha := \left(\bigcup_{0 \leq \beta < \alpha} \mathcal{E}_\beta \right)^*$$

and define

$$\mathcal{A} := \bigcup_{0 \leq \alpha < \Omega} \mathcal{E}_\alpha.$$

3. Show that each $\mathcal{E}_\alpha \subseteq \mathcal{B}$.

4. Show that $\mathcal{A} \subset \mathcal{B}$.
5. Now show that $\mathcal{A}$ is a σ -algebra to conclude that $\mathcal{A} = \mathcal{B}$.
 - (a) Show that $\emptyset, X \in \mathcal{A}$.
 - (b) Let $A \in \mathcal{A} \implies A \in \mathcal{E}_\alpha$ for some $\alpha < \Omega$. Show that this implies $\tilde{A} \in \mathcal{E}_\alpha^* \subset \mathcal{E}_\beta$ for every $\beta > \alpha$.
 - (c) Conclude that $\tilde{A} \in \mathcal{A}$ and thus conclude that $\mathcal{A}$ is closed under complementation.
 - (d) Now let $\{A_k\}$ be a sequence in $\mathcal{A}$. Show that $\bigcup_{k=1}^{\infty} A_k \in \mathcal{A}$. (Hint: Each A_k is in $\mathcal{A}_{\alpha_k}$ for some $\alpha_k < \Omega$. We know that there is $\beta < \Omega$ such that $\beta > \alpha_k$ for each $k \in \mathbb{N}$.)

Proof. 3. Since $\mathcal{E}_0 \subseteq \mathcal{B}$, only need to show that if $\mathcal{E}_\beta \subseteq \mathcal{B}$ for all $0 \leq \beta < \alpha$, then $\mathcal{E}_\alpha \subseteq \mathcal{B}$. Take any $E \in \mathcal{E}_\alpha$, then there are $E_1, \dots, E_k, \dots$, such that $E_i \in \bigcup_{0 \leq \beta < \alpha} \mathcal{E}_\beta$, or $E_i^c \in \bigcup_{0 \leq \beta < \alpha} \mathcal{E}_\beta$, and $E = \bigcup_{k=1}^{\infty} E_k$. We suppose that $E_i \in \mathcal{E}_{\beta_i}$ or $E_i^c \in \mathcal{E}_{\beta_i}$. Since $\mathcal{E}_{\beta_i} \subseteq \mathcal{B}$ for all β_i , and since $\mathcal{B}$ is closed under complement, then $E_i \in \mathcal{B}$. Since $\mathcal{B}$ is closed under countable union, $E = \bigcup_{k=1}^{\infty} E_k \in \mathcal{B}$. Thus $\mathcal{E}_\alpha \subseteq \mathcal{B}$.

4. We have shown that $\mathcal{E}_\alpha \subseteq \mathcal{B}$ for all $\alpha < \Omega$, then it is immediate that $\mathcal{A} \subseteq \mathcal{B}$.

5. (a) Since $\emptyset \in \mathcal{E}_0$ and $X = (\emptyset)^c \in \mathcal{E}_1 = (\mathcal{E}_0)^*$, then $\emptyset, X \in \mathcal{A}$.

(b) Write

$$\mathcal{E}_\alpha^* = \left\{ \bigcup_{k=1}^{\infty} E_k : \text{where either } E_i \in \mathcal{E}_\alpha \text{ or } E_i^c \in \mathcal{E}_\alpha \text{ for all } i \in \mathbb{N} \right\}$$

We take $E_1 = A^c$, then $E_1^c = A \in \mathcal{E}_\alpha$, and take $E_2 = \dots = \emptyset$. Then $A^c = \bigcup_{k=1}^{\infty} E_k \in \mathcal{E}_\alpha^*$. If $\beta > \alpha$, then

$$\mathcal{E}_\alpha \subseteq \bigcup_{0 \leq \gamma < \beta} \mathcal{E}_\gamma$$

Let $*$ act on both side, then it follows from the definition of $*$ that

$$(\mathcal{E}_\alpha)^* \subseteq \left(\bigcup_{0 \leq \gamma < \beta} \mathcal{E}_\gamma \right)^* = \mathcal{E}_\beta$$

Thus $A^c \in \mathcal{E}_\alpha^* \subseteq \mathcal{E}_\beta$ for every $\beta > \alpha$.

(c) It is immediate from (b) that $A^c \in \mathcal{E}_\beta \subseteq \mathcal{A}$.

(d) Suppose that $A_k \in \mathcal{E}_{\alpha_k}$ for some $\alpha_k < \Omega$. Let

$$\gamma := \bigcup_{k \in \mathbb{N}} \alpha_k, \quad \beta := \gamma + 1$$

Then $\gamma, \beta < \Omega$, and $\alpha_1, \dots, \alpha_k < \beta$. Then

$$\bigcup_{k=1}^{\infty} A_k \in \left(\bigcup_{k=1}^{\infty} \mathcal{E}_{\alpha_k} \right)^* \subseteq \mathcal{E}_\beta \subseteq \mathcal{A}$$

□

Exercise 1.2.7

An outer measure φ on a space X is called σ -finite if there exists a countable number of sets A_i with $\varphi(A_i) < \infty$ such that $X \subseteq \bigcup_{i=1}^{\infty} A_i$. Assuming that φ is a σ -finite Borel regular outer measure on a metric space X , prove that $E \subseteq X$ is φ -measurable if and only if there exists an F_σ set $F \subseteq E$ such that $\varphi(E \setminus F) = 0$.

Proof. Let $E_i := E \cap A_i$. Then

$$E = \bigcup_{i=1}^{\infty} E_i, \quad \varphi(E_i) \leq \varphi(A_i) < \infty$$

Once we show that there is a closed set $F_i \subseteq E_i$, such that $\varphi(E_i \setminus F_i) = 0$ for all i , then let $F = \bigcup_{i=1}^{\infty} F_i \in F_\sigma$, it follows that

$$\varphi(E \setminus F) \leq \varphi\left(\bigcup (E \setminus F_i)\right) \leq \varphi\left(\bigcup (E_i \setminus F_i)\right) \leq \sum_i \varphi(E_i \setminus F_i) = 0$$

□